

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech Examination (IT)

Apr-May 2018

IT247: Data Structures & Algorithms / IT214: Data Structures & Files

Date: 07.05.2018, Monday

Time: 10:00 am to 1:00 pm

Maximum Marks:70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Do as directed.

[07]

(1)

01

A certain sorting technique was applied to the following data set.

45, 54, 27, 36, 1, 90

After two passes, the rearrangement of the data set is given as below:

1, 27, 54, 36, 45, 90

Identify the sorting algorithm that was applied.

(2) Differentiate between DFS and BFS in graph with suitable example.

02

(3) The data structure used in hierarchical data model is

01

(a) Array (b) Linked list (c) Tree (d) Graph

(4) Define Data structure. Discuss the importance of data structure in computer field.

03

Q-2

[14]

(a) Search 09 in the given data: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.

04

Using (1) Linear Search (2) Binary Search. Calculate number of comparisons for both the search algorithms.

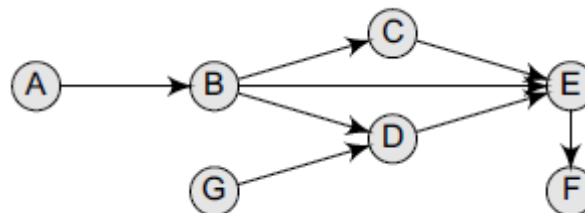
(b) Answer following questions.(Any Two)

10

1 Sort the following sequence of numbers in descending order using heap sort.

15, 19, 10, 7, 17, 16.

2 Consider a directed acyclic graph G given in below Figure



Using above dependency graph find the correct order of using topological sorting algorithm. Start with node A.

3 **Create Binary tree from following order.**

Inorder : G, D, H, L, B, E, A, C, I, F, K, and J

Postorder : G, L, H, D, E, B, I, K, J, F, C, and A

Also write Pre Order traversal sequence from generated tree.

- Q-3 Attempt Any Two.** [14]
- (a) Construct an AVL tree by inserting the following elements in the given order.
63, 9, 19, 27, 18, 108, 99, 81.
- (b) What is the basic idea of insertion sort? Apply insertion sort on below numbers.
Input Sequence : 39,9,45,63,18,81,108,54,72,36
What is the advantage of insertion sort compare to selection sort?
- (c) How does the height of a binary search tree affect its performance?
Create a binary search tree using the following data elements. Show the binary search tree after every step.
45, 39, 56, 12, 34, 78, 32, 10, 89, 54, 67, 81

SECTION – II

- Q-4 Do as directed** [7]
- 1 Consider the following operation performed on a stack of size 5. [01]
Push(1), Pop(), Push(2), Push(3), Pop(), Push(4), Pop(), Pop(), Push(5), Push(3)
After the completion of all operation, How many elements are present in stack?
- 2 Which data structure is used during Breadth First Traversal on a graph? [01]
- 3 Which data structure is used to implement chaining technique for resolving hash collision in hash table? [01]
- 4 Define input restricted Deque. [01]
- 5 What are the overflow and underflow conditions for Circular queue? Explain with suitable example. [03]
- Q-5** [14]
- (a) Generate index for each data using hash function and arrange them into an array of seven elements. Use linear probing to resolve collision. Array index starts at 0. [04]
Hash function $H(x) = X \bmod 4$, DATA: 65, 74, 71, 66, 80, 90
- (b) **Answer following questions.(Any Two)** [10]
- (1) Convert following notation into postfix notation. Show each operation using stack.
 $a + (((b - c) * (d - e) + f) / g) ^ (h - j)$
- (2) Write an algorithm or pseudo code for following operations in a doubly linked list.
1) count number of nodes
2) print nodes in reverse order
- (3) What is collision in hashing? Explain any one collision-resolution technique with suitable example.
- Q-6** [14]
- (a) For given 3D array A [-2:0,1: 4,6:9] of integer elements. Find the total number of elements. Also find address of A [-1, 3,8] element in column major order. Base address is 2000. [04]
- (b) **Answer following questions.(Any Two)** [10]
- 1 Write pseudo code or steps to delete specific node of the given singly linked list. Consider all the cases.
- 2 Define priority queue. List out different methods to implement priority queue. Explain any one method to implement priority queue.
- 3 Evaluate following postfix notation using stack.
 $3\ 4 + 5\ 2 / 2 * 2 / 2 / - 9 +$